Algorithm and Data Structures I. example Test

1.) Run the mergesort algorithm on the array A[15; 14; 55; 7; 9; 49; 16; 21; 19]. List all of the nested recursive calls with the appropriate parameters as well as the rearrangements within them.

2.) We are given two increasingly sorted arrays $B[1 : k]$ and $C[1 : l]$ without repetition. Implement a procedure (structogram) that produces an increasingly sorted array A containing the elements of the symmetric difference of B and C. The algorithm should use not more than $k + l$ comparisons.

3.) Given a string (up to n characters). Using a stack, decide that the input string is a palindrome. (Reading back and forth is the same.)

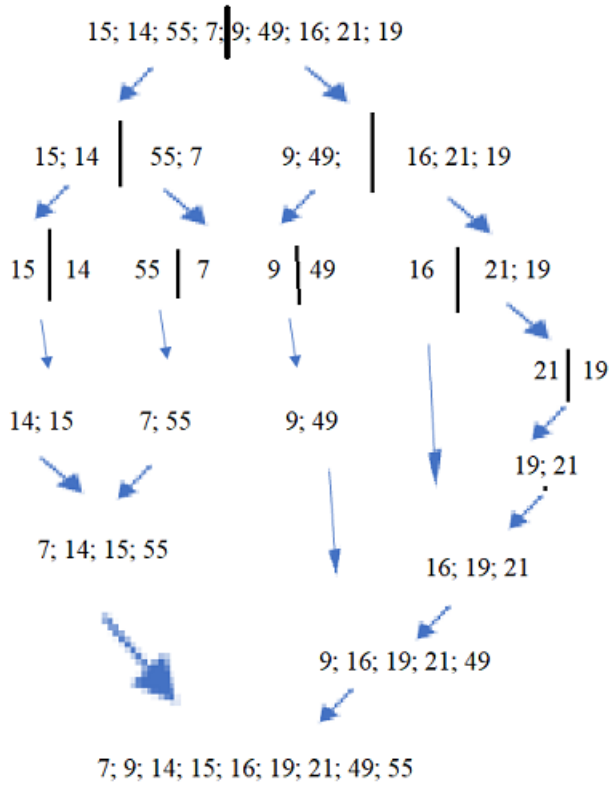
4.) Convert the following (infix) algebraic expression to postfix expression (Reverse Polish Notation) using the shunting-yard algorithm explained in the practice:

```
x=a*(b+(5*y)*h^4^g)/i+y^z+1*t+e
```

5.) Pointer H refers to the header of an unsorted **H1L**. Write function **remove_max(*H*)** which removes an element with maximal key from the list and returns the address of the element removed. $T(n) \in \Theta(n)$ where n is the length of the list.

Solution of the sample test

1)



2)

SymmetricalDifference(B/1:T[k],C/1:T[l]:A/1:T[k+l])

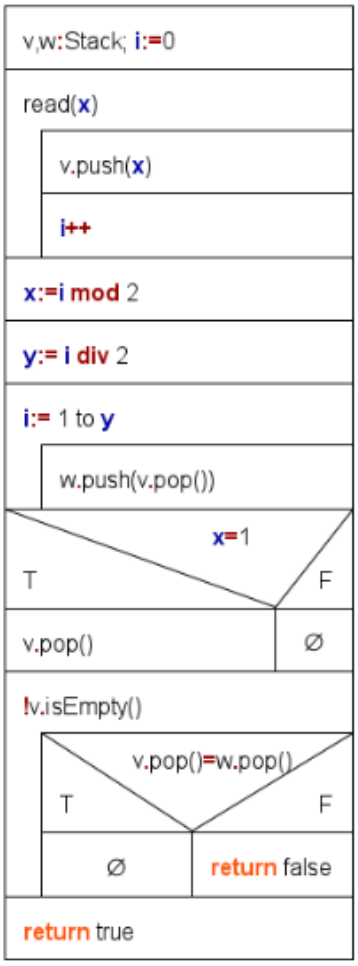
i:=1; j:=1; m:=0
i≤k **and** j≤l

(?)		
B[i] < C[j]	B[i] = C[j]	B[i] > C[j]
m:=m+1	i:=i+1	m:=m+1
A[m]:=B[i]	j:=j+1	A[m]:=C[j]
i:=i+1		j:=j+1

i≤k
m:=m+1
A[m]:=B[i]
i:=i+1
j≤l
m:=m+1
A[m]:=C[j]
j:=j+1

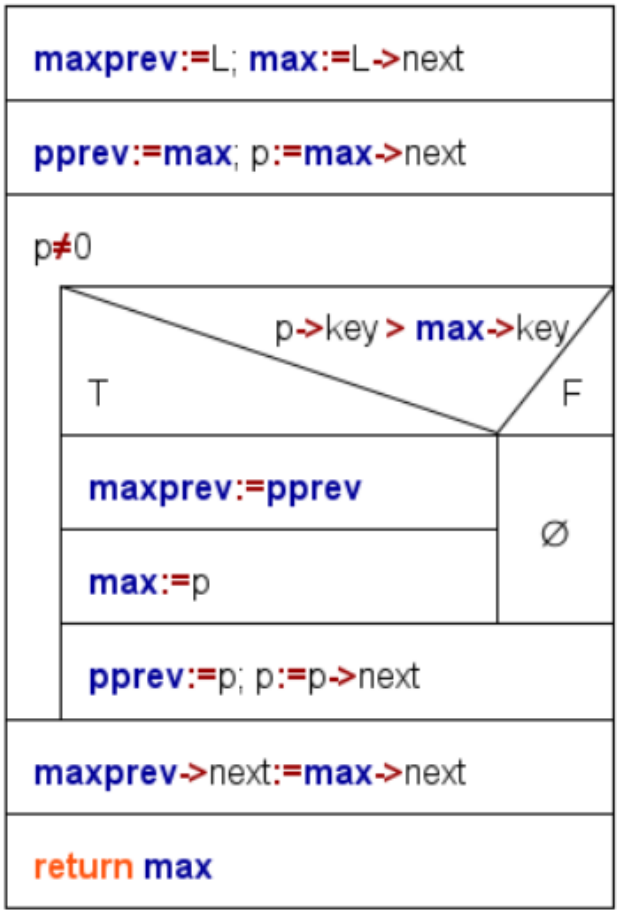
3)

Palindrom():B



4)

Remove_Max(H:E1*):E1*



5)

